

# Operations Manual

## Safaricom Ethiopia SDP

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## Revision History

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## 1 Introduction

The purpose of this document is to describe operational tasks with steps for Safaricom. Following are the modules involved in this deployment.

- Subscription Management

This manual is intended for anyone who is acting as an administrator for Six DEE's MVNE product maintenance.

## 2 Subscription Management

Following are the Kubernetes cluster Jump server servers where Subscription Management is installed.

| Server IP   | Instance |
|-------------|----------|
| 10.3.24.21  | EDGE DC1 |
| 10.3.124.21 | EDGE DC2 |

### 2.1 Application Yaml and config Path

Below are the Application deployment files path in jump Server.

| Micro Service Name                           | Application Path                                    | Config Path                                         | Server |
|----------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|--------|
| sdp-act<br>sdp-consent<br>sdp-requestsms     | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp | DC1    |
| sdp-payment                                  | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp |        |
| sdp-sendsms                                  | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp |        |
| sdp-deactivation<br>sdp-usgrpt<br>sdp-preren | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp |        |
| sdp-ren<br>sdp-mtcon<br>sdp-delrcpt          | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp |        |

|                    |                                                                    |                                                                    |     |
|--------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|-----|
| sdp-mprofile       | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp                | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp                | DC1 |
| Sdp-esme           | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp                | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp                | DC1 |
| Sdp-morouter       | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp                | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp                | DC1 |
| cms-upc-web-portal | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/cms/cms-upc-portal | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/cms/cms-upc-portal | DC1 |

| Micro Service Name                           | Application Path                                    | Config Path                                         | Server |
|----------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|--------|
| sdp-act<br>sdp-consent<br>sdp-requestsms     | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2    |
| sdp-payment                                  | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2    |
| sdp-sendsms                                  | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2    |
| sdp-deactivation<br>sdp-usgrpt<br>sdp-preren | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2    |

|                                     |                                                     |                                                     |     |
|-------------------------------------|-----------------------------------------------------|-----------------------------------------------------|-----|
| sdp-ren<br>sdp-mtcon<br>sdp-delrcpt | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2 |
| sdp-mprofile                        | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2 |
| Sdp-esme                            | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2 |
| Sdp-morouter                        | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | DC2 |

|                    |                                                                    |                                                                    |     |
|--------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|-----|
| cms-upc-web-portal | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/cms/cms-upc-portal | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/cms/cms-upc-portal | DC2 |
|--------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|-----|

## 2.2 Logs Path

Below are the Log paths for SDP System. These paths are the same for both DC1 and DC2.

To access the paths below, first we need to login to the pod of the application by using the below command.

**Command:** `kubectl exec -it {pod name} -c {container name} -sh`

| Pod Name  | Container Name                           | Application Logs Path | File Name                          |
|-----------|------------------------------------------|-----------------------|------------------------------------|
| Sdp-pod-1 | sdp-act<br>sdp-consent<br>sdp-requestsms | /data/log             | {podname}-<br>{containername}*.log |
| Sdp-pod-2 | sdp-payment                              | /data/log             | {podname}-<br>{containername}*.log |

|              |                                              |           |                                    |
|--------------|----------------------------------------------|-----------|------------------------------------|
| Sdp-pod-3    | sdp-sendsms                                  | /data/log | {podname}-<br>{containername}*.log |
| Sdp-pod-4    | sdp-deactivation<br>sdp-usgrpt<br>sdp-preren | /data/log | {podname}-<br>{containername}*.log |
| Sdp-pod-5    | sdp-ren<br>sdp-mtcon<br>sdp-delrcpt          | /data/log | {podname}-<br>{containername}*.log |
| Sdp-pod-6    | sdp-mprofile                                 | /data/log | {podname}-<br>{containername}*.log |
| Sdp-esme     | Sdp-esme                                     | /data/log | {podname}-<br>{containername}*.log |
| Sdp-morouter | Sdp-morouter                                 | /data/log | {podname}-<br>{containername}*.log |

## 2.3 EDR Path

Below are the EDR paths for Billing System. These paths are the same for both DC1 and DC2.

To access the paths below, first we need to login to the pod of the application by using the below command.

**Command:** `kubect exec -it {pod name} – c {container name} -- sh`

| Pod Name  | Container Name                           | Application Logs Path | Topic Name |
|-----------|------------------------------------------|-----------------------|------------|
| Sdp-pod-1 | sdp-act<br>sdp-consent<br>sdp-requestsms | /data/cdr             | 6d-sdp-cdr |
| Sdp-pod-2 | sdp-payment                              | /data/cdr             | 6d-sdp-cdr |
| Sdp-pod-3 | sdp-sendsms                              | /data/cdr             | 6d-sdp-cdr |

|           |                                              |           |             |
|-----------|----------------------------------------------|-----------|-------------|
| Sdp-pod-4 | sdp-deactivation<br>sdp-usgrpt<br>sdp-preren | /data/cdr | 6d-sdp-cdr  |
| Sdp-pod-5 | sdp-ren<br>sdp-mtcon<br>sdp-delrcpt          | /data/cdr | {6d-sdp-cdr |
| Sdp-pod-6 | sdp-mprofile                                 | /data/cdr | 6d-sdp-cdr  |

## 2.4 Application Status

Execute the following command to check the Billing Pods status.

| Instance Name           | Command                       |
|-------------------------|-------------------------------|
| Subscription Management | kubectl get pods   grep "sdp" |

If the application is running, then the following output is displayed

### Edge DC 1 Jump Server (10.3.24.21):

```
[anila.pyreddy@et01-bss-jump01 ~]$ kubectl get po | grep sdp
apigw-sdp-pod-1-59648dd484-lmclq      2/2      Running    0          20h
sdp-esme-859db59756-6ws5x           1/1      Running    0          5d9h
sdp-morouter-7b8476cf55-txcdh        1/1      Running    0          5d9h
sdp-morouter-proxysql-8cb8fcdcb-7kzcd 1/1      Running    0          31d
sdp-morouter-proxysql-8cb8fcdcb-ltsss 1/1      Running    1          43d
sdp-pod-1-5c555469bd-75n5n          6/6      Running    0          5d9h
sdp-pod-2-74d55cfb4f-7bvlp          4/4      Running    0          5d9h
sdp-pod-3-559b88f68d-wbdxx          4/4      Running    0          5d9h
sdp-pod-4-66dbf6c87d-qvnrq          6/6      Running    1          4d15h
sdp-pod-5-77ddb4576-q5t8s           7/7      Running    0          5d9h
sdp-pod-6-7897b5dddf-pwjb4          4/4      Running    0          4d15h
sdp-proxysql-1-77476d64f9-88h42      1/1      Running    1          43d
sdp-proxysql-1-77476d64f9-tlkwj      1/1      Running    0          27d
sdp-proxysql-2-c75996d5f-kp4gq       1/1      Running    1          43d
sdp-proxysql-3-5ff576f59b-6ws4       1/1      Running    1          43d
sdp-proxysql-4-768cc9ddc6-kqdrq      1/1      Running    0          26d
sdp-proxysql-4-768cc9ddc6-vm8wj       1/1      Running    1          43d
sdp-proxysql-5-5974cb8bb8-psxgc       1/1      Running    1          43d
sdp-proxysql-5-5974cb8bb8-zsssg       1/1      Running    0          27d
sdp-proxysql-6-7df5dd8845-jgnnh       1/1      Running    1          43d
sdp-upc-proxysql-5bdb4746b6-ss457     1/1      Running    0          6d2h
sdp-upc-proxysql-5bdb4746b6-zh2ll     1/1      Running    1          43d
```

### Edge DC 2 Jump Server (10.3.124.21):

```
[anila.pyreddy@et01-bss-jump01 ~]$ kubectl get po | grep sdp
apigw-sdp-pod-1-59648dd484-lmclq      2/2      Running    0          20h
sdp-esme-859db59756-6ws5x           1/1      Running    0          5d9h
sdp-morouter-7b8476cf55-txcdh        1/1      Running    0          5d9h
sdp-morouter-proxysql-8cb8fcdcb-7kzcd 1/1      Running    0          31d
sdp-morouter-proxysql-8cb8fcdcb-ltsss 1/1      Running    1          43d
sdp-pod-1-5c555469bd-75n5n          6/6      Running    0          5d9h
sdp-pod-2-74d55cfb4f-7bvlp          4/4      Running    0          5d9h
sdp-pod-3-559b88f68d-wbdxx          4/4      Running    0          5d9h
sdp-pod-4-66dbf6c87d-qvnrq          6/6      Running    1          4d15h
sdp-pod-5-77ddb4576-q5t8s           7/7      Running    0          5d9h
sdp-pod-6-7897b5dddf-pwjb4          4/4      Running    0          4d15h
sdp-proxysql-1-77476d64f9-88h42      1/1      Running    1          43d
sdp-proxysql-1-77476d64f9-tlkwj      1/1      Running    0          27d
sdp-proxysql-2-c75996d5f-kp4gq       1/1      Running    1          43d
sdp-proxysql-3-5ff576f59b-6ws4       1/1      Running    1          43d
sdp-proxysql-4-768cc9ddc6-kqdrq      1/1      Running    0          26d
sdp-proxysql-4-768cc9ddc6-vm8wj       1/1      Running    1          43d
sdp-proxysql-5-5974cb8bb8-psxgc       1/1      Running    1          43d
sdp-proxysql-5-5974cb8bb8-zsssg       1/1      Running    0          27d
sdp-proxysql-6-7df5dd8845-jgnnh       1/1      Running    1          43d
sdp-upc-proxysql-5bdb4746b6-ss457     1/1      Running    0          6d2h
sdp-upc-proxysql-5bdb4746b6-zh2ll     1/1      Running    1          43d
[anila.pyreddy@et01-bss-jump01 ~]$
[anila.p@et02-bss-jump01 SM]$ k get po | grep sm-
sm-pod-1-6b48d96ffc-rbkq5             7/7      Running    0          5d1h
[anila.p@et02-bss-jump01 SM]$
```

## 2.5 Start, Stop and Restart Billing Pods

Execute following commands to start, stop, and to restart the Billing System pods in DC1.

| Activity | folder Path                                         | Command                                  |
|----------|-----------------------------------------------------|------------------------------------------|
| Start    | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | Kubectllapply -k .                       |
| Stop     | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | Kubectl apply -k .                       |
| Restart  | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp | kubectl rollout restart {deploymentname} |

## 2.6 DB Details Billing System

Following are the various configuration paths with filename in ARM.

| Parameter Name | Property file and path | Tag |
|----------------|------------------------|-----|
|----------------|------------------------|-----|

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC1 | <p>/opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-1/proxy-pod-configs</p> <p>/opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-2/proxy-pod-configs</p> <p>/opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-3/proxy-pod-configs</p> <p>/opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-4/proxy-pod-configs</p> <p>/opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-5/proxy-pod-configs</p> | <p>jdbc:mysql://localhost:6033/WKN_SDP</p> <p>jdbc:mysql://localhost:6033/WKN_MOR</p> <pre>mysql_servers = (   { address="10.3.83.244" , port=3306 , hostgroup=1, max_connections=780,max_re   { address="10.3.83.245" , port=3306 , hostgroup=1, max_connections=780,max_re   { address="10.3.83.251" , port=3306 , hostgroup=1, max_connections=780,max_re   { address="10.3.183.243" , port=3306 , hostgroup=1, max_connections=780,max_r   { address="10.3.183.249" , port=3306 , hostgroup=1, max_connections=780,max_r )</pre> |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|     |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | /opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-6/proxy-pod-configs<br><br>/opt/safaricom-wakanda/overlay/edge-dc1-cluster/sdp/proxysql-pod-configs/sdp-morouter-proxysql/proxy-pod-configs                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DC2 | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-1/proxy-pod-configs<br><br>/opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-2/proxy-pod-configs<br><br>/opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-3/proxy-pod-configs | jdbc:mysql://localhost:6033/WKN_SDP<br><br>jdbc:mysql://localhost:6033/WKN_MOR<br><br><pre>mysql_servers = (   { address="10.3.83.244" , port=3306 , hostgroup=1, max_connections=780,     { address="10.3.83.245" , port=3306 , hostgroup=1, max_connections=780,     { address="10.3.83.251" , port=3306 , hostgroup=1, max_connections=780,     { address="10.3.183.243" , port=3306 , hostgroup=1, max_connections=780,     { address="10.3.183.249" , port=3306 , hostgroup=1, max_connections=780   )</pre> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | /opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-4/proxy-pod-configs<br><br>/opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-5/proxy-pod-configs<br><br>/opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-proxysql-pod-6/proxy-pod-configs<br><br>/opt/safaricom-wakanda/overlay/edge-dc2-cluster/sdp/proxysql-pod-configs/sdp-morouter-proxysql/proxy-pod-configs |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

## 2.5 Subscription Management URLs

Following are the URLs to access the Subscription Management

| Services   | URL                               | Server |
|------------|-----------------------------------|--------|
| Activation | http://sdp-pod-1:3081/Activation' |        |

|                   |                                          |              |
|-------------------|------------------------------------------|--------------|
| Payment           | http://sdp-pod-2:3081/Payment'           | DC1 /D<br>C2 |
| Activate          | 'http://sdp-pod-3:3081/SendSMS'          |              |
| Deactivate        | 'http://sdp-pod-4:3081/Deactivation'     |              |
| View Transactions | 'http://sdp-pod-4:3084/Transaction/View' |              |
| UpdateLanguage    | 'http://sdp-pod-6:3081/LanguageInfo'     |              |
| Deactivation ALL  | 'http://sdp-pod-4:3081/DeactivationALL'  |              |